mhs-poisson-ltau: magnetohydrostatic correction on an optical-depth grid

Physics and method of `run.mode = "mhs-poisson-ltau"` in HSEMHS. A self-contained description.

Abstract

An inversion of solar spectra gives the temperature and the magnetic field on surfaces of constant optical depth. This mode computes the gas pressure on every optical-depth surface *and* the geometric height of every surface in every pixel column such that pressure, gravity and the Lorentz force of the given field balance as well as possible in all three directions. On an optical-depth grid the two cannot be separated: the heights of the surfaces follow from the opacity, which depends on the pressure. We derive the vertical and horizontal force balance on tilted optical-depth surfaces, the Lorentz force in these curvilinear coordinates, and the least-squares objective; we then describe the Gauss–Newton (Levenberg–Marquardt) iteration with its exact linearisation, the preconditioner, the step control and the stopping rule. We discuss what fixes the pressure level (no external reference pressure is used), what fixes the heights, and which quantities remain undetermined. All assumptions are listed.

1. Introduction

A spectral line inversion describes the atmosphere in each pixel as a function of the continuum optical depth at 500 nm, τ_{500}: it gives T and $\mathbf{B}$ at, e.g., $\log_{10}\tau_{500} = -8.0, -7.9, \ldots, +1.4$. The simplest way to give such an atmosphere a pressure and a geometry is to put every pixel column into hydrostatic equilibrium on its own (the mode `column-hse-ltau`). The columns are then not in horizontal force balance with each other, and the Lorentz force of the field is ignored. This mode goes further, in the spirit of a magnetohydrostatic "Poisson step" on a geometric grid: T and $\mathbf{B}$ on the τ surfaces stay fixed, and we look for the pressure on every τ surface and the height of every τ surface in every column that together balance the forces best.

Input: the $\log_{10}\tau_{500}$ levels; the horizontal coordinates x, y (periodic box); T, B_x, B_y, B_z on every level of every column; the gravity g. **Output:** on the same τ levels, P, ρ and the height z; the height of $\tau_{500} = 1$ per column. T and $\mathbf{B}$ are copied unchanged.

2. Notation

Symbol	Meaning	Unit	Role
$x,\ y$	horizontal coordinates, periodic; x_i, y_j, spacing Δx, Δy	m	given
$\tau,\ \ell$	optical depth at 500 nm; $\ell = \log_{10}\tau$	–	coordinate
ℓ_k	levels $k = 1 \ldots N_\tau$, increasing (top first), $\Delta\ell = 0.1$	–	given
s	$\ln\tau = \ell \ln 10$; $\Delta s_k = s_{k+1} - s_k$	–	coordinate
$T,\ \mathbf{B}$	temperature and magnetic field on the levels	K, T	given, fixed
$P,\ u$	gas pressure on the levels; $u = \ln(P/1\,\mathrm{Pa})$	Pa, –	unknown below the cut
z_{ijk}	height of level k in column (i, j) (upward)	m	derived from u and c
$k_c,\ \ell_{\mathrm{cut}}$	cut level: the level nearest to ℓ_{cut} (`logtau_cut`, default -4)	–	parameter
c_{ij}	height of the cut level in column (i, j)	m	unknown
n	number of solved levels, $n = N_\tau - k_c + 1$	–	derived
$\rho,\ \chi$	density and 500 nm extinction, $\rho(T, P)$, $\chi(T, P)$	$\mathrm{kg\,m^{-3}}$, $\mathrm{m^{-1}}$	derived
$q,\ a$	$q = \rho/P$; $a = \tau/\chi$ (thickness per unit $\ln\tau$)	$\mathrm{s^2\,m^{-2}}$, m	derived
χ'	$\partial \ln\chi/\partial \ln P$ at fixed T	–	derived
$\mathbf{J},\ \mathbf{L}$	current $\nabla\times\mathbf{B}/\mu_0$ and Lorentz force $\mathbf{J}\times\mathbf{B}$ in physical space	$\mathrm{A\,m^{-2}}$, $\mathrm{N\,m^{-3}}$	derived from $\mathbf{B}$ and z
W	$(\rho g - L_z)/P = g\,q - L_z e^{-u}$: effective weight per unit pressure	$\mathrm{m^{-1}}$	derived
$\mathbf{F},\ \mathbf{R}$	net force per volume; relative residual $\mathbf{F}/P$	$\mathrm{N\,m^{-3}}$, $\mathrm{m^{-1}}$	residual
$\bar{H}_k,\ \overline{\Delta z}_k$	layer medians of the scale height $P/(\rho g)$ and of the layer thickness in the starting state	m	fixed weights
Φ	objective (weighted sum of squared residuals)		minimised
μ_{LM}	Levenberg–Marquardt damping	–	iteration
$g,\ \mu_0$	gravity 274; vacuum permeability $4\pi\times10^{-7}$	$\mathrm{m\,s^{-2}}$, $\mathrm{T\,m\,A^{-1}}$	constants

3. The force balance

A static gas has zero net force per unit volume:

$$\mathbf{F} = -\nabla P + \mathbf{J} \times \mathbf{B} - \rho g\, \hat{\mathbf{z}} = 0, \qquad \mathbf{J} = \frac{\nabla \times \mathbf{B}}{\mu_0}, \qquad \mathbf{J} \times \mathbf{B} = -\nabla\left(\frac{B^2}{2\mu_0}\right) + \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mu_0}. \tag{1}$$

The Lorentz force is a magnetic pressure plus a tension along curved field lines. With the field fixed this balance can in general not be satisfied exactly. Taking the curl of Eq. (1) removes the pressure and leaves $\nabla \times (\mathbf{J} \times \mathbf{B}) = g\,(\partial_y \rho, -\partial_x \rho, 0)$: only the tension of the field can hold up horizontal density differences, and a field taken from an inversion has no reason to have exactly the right tension. We therefore look for the state that balances the forces *as well as possible* (Sect. 9).

The new difficulty here is that all derivatives in $\mathbf{F}$ are taken in physical space, at fixed height, whereas the data live on τ surfaces whose heights are themselves unknown.

4. Optical depth, heights and mass

4.1 Heights follow from the opacity

Going down by dz adds the optical depth $d\tau = -\chi\, dz$. With $s = \ln \tau$,

$$\frac{dz}{ds} = -\frac{\tau}{\chi(T,P)} = -a, \qquad z_{k+1} = z_k - \int_{s_k}^{s_{k+1}} a\, ds \approx z_k - \frac{\Delta s_k}{2}\,(a_k + a_{k+1}). \tag{2}$$

The height of a level is the height of the level above it minus the thickness of the layer in between, and the thickness is set by the opacity, which depends on the pressure. In a column, the heights below the cut level k_c are therefore

$$\boxed{z_{k_c} = c, \qquad z_k = c - \sum_{m=k_c}^{k-1} \frac{\Delta s_m}{2}\,(a_m + a_{m+1}), \quad k > k_c} \tag{3}$$

Two consequences: *a change of pressure at one level moves every level below it* (higher pressure, more opacity, a thinner layer, everything below shifts up); and the unknowns of a column can be taken to be the pressures on its levels plus one height c. All other heights follow. Equation (3) is a cumulative sum: a lower-triangular linear operator acting down the column.

4.2 Optical depth measures mass

Since $\chi = \kappa\rho$ with κ the extinction per unit mass and $dm = -\rho\, dz$ the mass added going down, $d\tau = \kappa\, dm$: optical depth is column mass weighted by the opacity. This is why, unlike on a geometric grid, the pressure on a τ grid needs no external reference value (Sect. 10.3).

5. The equation of state and the opacity

Every force balance in this document involves the density ρ, and every optical depth involves the extinction coefficient χ_{500}. Neither is an unknown of its own: at given temperature and pressure both are fixed by the microphysics of the gas. This section explains how.

5.1 The ideal gas

In the solar photosphere and chromosphere the particles interact only weakly, so the gas obeys the ideal-gas law. If n is the number of free particles (atoms, ions, molecules and electrons) per cubic metre,

$$P = n\, k_B T. \tag{4}$$

If the average mass of a free particle is $\bar{\mu}\, m_\mathrm{u}$ (with m_u the atomic mass unit and $\bar{\mu}$ the **mean molecular weight**), the mass density is $\rho = n\,\bar{\mu}\, m_\mathrm{u}$, and therefore

$$\rho = \frac{\bar{\mu}\, m_\mathrm{u}}{k_B}\,\frac{P}{T}. \tag{5}$$

At fixed $\bar{\mu}$ and T the density is proportional to the pressure.

5.2 Why $\bar{\mu}$ depends on T and P: ionisation and molecules

For neutral solar gas (about 91 % hydrogen and 9 % helium atoms by number, plus a small fraction of heavier elements) $\bar{\mu} \approx 1.3$. When a hydrogen atom is ionised it becomes two free particles, a proton and an electron, of

almost the same total mass; in fully ionised gas $\bar{\mu} \approx 0.6$. In the coolest layers hydrogen partly forms H_2 molecules, which raises $\bar{\mu}$. The degree of ionisation of an element in stage i is given by the **Saha equation**

$$\frac{n_{i+1}\, n_e}{n_i} = \frac{2U_{i+1}(T)}{U_i(T)} \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} e^{-E_i/k_B T}, \tag{6}$$

with E_i the ionisation energy (13.6 eV for hydrogen), U_i the partition functions, m_e the electron mass and h Planck's constant. Two properties matter for us. (i) Because of the exponential, ionisation switches on over a narrow range of temperature. (ii) The electron density n_e appears on the left: at higher density there are more electrons to recombine with, and a higher temperature is needed for the same ionisation. Hence $\bar{\mu} = \bar{\mu}(T, P)$, and Eq. (5) is no longer a simple proportionality. The Saha equations of all elements are coupled through the common n_e and through charge conservation (n_e equals the total charge of the ions), and the molecular equilibria (H_2, H_2^+, H^-) are solved together with them. This assumes **local thermodynamic equilibrium** (LTE): the level populations and ionisation stages are those of a gas in equilibrium at the local T.

5.3 The 500 nm extinction

Light of wavelength 500 nm travelling a distance ds loses the fraction $\chi_{500}\, ds$ of its intensity, by absorption and by scattering. In the solar photosphere χ_{500} is dominated by the negative hydrogen ion H^- (a hydrogen atom with a second, weakly bound electron, binding energy 0.75 eV), whose abundance is proportional to the electron density. The electrons come mostly from easily ionised metals (Na, Mg, Si, Ca, Fe, ...), so χ_{500} grows steeply with temperature and with density. The code adds the continuum contributions of H (bound–free and free–free), H^-, H_2^+, He, He^-, the metals C, N, Mg, Si, Ca and Fe, Rayleigh scattering on H, He and H_2, and Thomson scattering on free electrons, each counted once. Spectral lines are not included: χ_{500} is a continuum extinction.

5.4 The table and its interpolation

Solving the coupled Saha, molecular and charge equations at every grid point at every step would be far too slow. They are therefore solved once, with Wittmann's equation-of-state code for solar composition (with a dedicated hot branch above a switch temperature $T_{\rm sw}$, where the original solver fails in the hot, low-pressure corner), on a grid of $X = \ln T$ and $Y = \ln P$, and stored in a table. For every node the table holds the three functions

$$\ln \rho(X,Y), \qquad \ln n_e(X,Y), \qquad \ln \chi_{500}(X,Y),$$

together with their derivatives ∂_X, ∂_Y and $\partial_X \partial_Y$. Between the nodes each function is evaluated by **bicubic Hermite interpolation**: a cubic polynomial in each direction that matches the stored values and derivatives at the four corners of the cell. The result is continuous with continuous first derivatives, which is important for the Newton-type solvers of the magnetohydrostatic modes, and it provides the derivative $\partial \ln \rho / \partial \ln P|_T$ that they need. The table used by all modes is `cache/eos_table_fullz.h5`. Its range of T and P (its *envelope*) covers the input atmospheres; a point outside the envelope is reported as an equation-of-state failure, never extrapolated.

Two derived quantities appear repeatedly:

$$q \equiv \frac{\rho}{P} \quad [\mathrm{s^2\,m^{-2}}], \qquad H \equiv \frac{P}{\rho\, g} = \frac{1}{g\, q} = \frac{k_B T}{\bar{\mu}\, m_{\rm u}\, g} \quad [\mathrm{m}]. \tag{7}$$

H is the **pressure scale height**. For $T = 6000$ K and $\bar{\mu} = 1.3$, $H \approx 140$ km.

6. Vertical balance along τ

The vertical component of Eq. (1) is $\partial P/\partial z = -(\rho g - L_z)$: the pressure difference across a slab carries its weight minus what the vertical Lorentz force carries. Divide by $dz = -a\, ds$ and by P:

$$\boxed{\frac{d\ln P}{d\ln \tau} = \frac{(\rho g - L_z)\,\tau}{\chi\, P} = W\, a}, \qquad W \equiv \frac{\rho g - L_z}{P} = g\, q - L_z\, e^{-u}. \tag{8}$$

Without the field this is the hydrostatic equation in optical depth, $dP/d\tau = g/\kappa$. An upward Lorentz force ($L_z > 0$) carries part of the weight, so the pressure rises less steeply with optical depth.

7. Horizontal balance on tilted surfaces

A τ surface is generally not flat: its height $z(x, y, \tau)$ varies across the box. Consider the pressure along such a surface, $\tilde{P}(x, \tau) = P\big(x, z(x, \tau)\big)$. By the chain rule,

$$\left.\frac{\partial P}{\partial x}\right|_\tau = \left.\frac{\partial P}{\partial x}\right|_z + \frac{\partial P}{\partial z}\, \left.\frac{\partial z}{\partial x}\right|_\tau . \tag{9}$$

The physical force needs $\partial P/\partial x$ at fixed height. Solving for it and using the vertical balance $\partial P/\partial z = -(\rho g - L_z)$:

$$\boxed{F_x = -\left.\frac{\partial P}{\partial x}\right|_\tau - (\rho g - L_z)\left.\frac{\partial z}{\partial x}\right|_\tau + L_x, \qquad F_y = -\left.\frac{\partial P}{\partial y}\right|_\tau - (\rho g - L_z)\left.\frac{\partial z}{\partial y}\right|_\tau + L_y} \tag{10}$$

The middle term has a simple meaning. If a τ surface slopes, comparing pressures along it means comparing points at different heights, and the weight of the gas between them must be subtracted: on a tilted surface part of the weight acts sideways. Horizontal balance therefore ties together the pressures *and* the heights of neighbouring columns: a pressure difference along a τ surface can be balanced by a tilt of the surface.

8. The Lorentz force on a τ grid

$\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$ needs derivatives in physical space. With the field given on the τ levels, the same chain rule gives, for any field component f,

$$\frac{\partial f}{\partial z} = \frac{\partial f/\partial s}{\partial z/\partial s}, \qquad \left.\frac{\partial f}{\partial x}\right|_z = \left.\frac{\partial f}{\partial x}\right|_\tau - \left.\frac{\partial z}{\partial x}\right|_\tau \frac{\partial f}{\partial z}, \qquad \left.\frac{\partial f}{\partial y}\right|_z = \left.\frac{\partial f}{\partial y}\right|_\tau - \left.\frac{\partial z}{\partial y}\right|_\tau \frac{\partial f}{\partial z}. \tag{11}$$

Then $\mathbf{J} = \mu_0^{-1}(\partial_y B_z - \partial_z B_y,\ \partial_z B_x - \partial_x B_z,\ \partial_x B_y - \partial_y B_x)$ and $\mathbf{L} = \mathbf{J} \times \mathbf{B}$. So even though the field values on the τ levels are fixed, the currents and the Lorentz force depend on where the levels lie: if two levels move apart, the vertical gradients of the field between them become smaller, and so do the currents they imply. The Lorentz force is therefore recomputed whenever the heights change, and the solver takes its dependence on the heights into account (Sect. 13.2).

9. The quantity minimised

9.1 Relative residuals

As on a geometric grid, the forces are divided by the pressure, so that lowering the pressure is not rewarded:

$$R_x = \frac{F_x}{P} = -\left.\frac{\partial u}{\partial x}\right|_\tau - W\left.\frac{\partial z}{\partial x}\right|_\tau + L_x e^{-u}, \qquad R_y = \frac{F_y}{P}\ \text{(likewise)}, \qquad R_z = \Delta u - \int W\, a\, ds. \tag{12}$$

R_x, R_y are evaluated on every solved level; R_z is the vertical balance (8) integrated over one level interval, so it is the misfit of $\ln P$ across the interval.

9.2 Least squares

At every point there are three equations; per column the unknowns are $n-1$ pressures and one height (Sect. 10). There are more equations than unknowns and, by Sect. 3, no exact solution. The pressure and heights are therefore those that minimise

$$\boxed{\Phi = \sum_{i,j}\left[\sum_{k=k_c}^{N_\tau} \overline{\Delta z}_k\, \bar{H}_k^2\left(R_{x,ijk}^2 + R_{y,ijk}^2\right) + \sum_{k=k_c}^{N_\tau - 1} \frac{\bar{H}_k \bar{H}_{k+1}}{\overline{\Delta z}_{k+1/2}}\, R_{z,ij,k+1/2}^2\right]} \tag{13}$$

$\bar{H}_k$ is the median over the box of the scale height $P/(\rho g)$ on level k and $\overline{\Delta z}_{k+1/2}$ the median thickness of interval k, both from the starting state and fixed during the solve; $\overline{\Delta z}_k$ is the mean of the two adjacent interval thicknesses (half the single one at the end levels). Since $\bar{H} R_x \approx F_x/(\rho g)$ and $\bar{H} R_z/\overline{\Delta z} \approx F_z/(\rho g)$, Φ approximates a volume integral (per unit area) of the squared imbalance in units of the weight of the gas, so that the dense deep layers do not swamp the upper ones.

10. The unknowns and what fixes the pressure level

10.1 Unknowns

In every column the solve determines

- the height c_{ij} of the cut level k_c (the level nearest to $\log_{10}\tau = -4$), and
- $u_{ijk} = \ln P$ on the levels $k = k_c + 1, \ldots, N_\tau$ below the cut.

All heights below the cut follow from Eq. (3). The total is n unknowns per column against $3n - 1$ residuals.

10.2 Above the cut

Above the cut the field is strong compared with the thin gas (low plasma β), and a field taken from an inversion is far from force-free there. Balancing its Lorentz force by the pressure would require enormous pressure gradients that would propagate down into the photosphere. Above the cut ($\log_{10}\tau < -4$) the levels are therefore not solved: their pressures and their distances to the cut level stay those of the starting state, which is hydrostatic in τ, and the whole upper part of a column moves rigidly up and down with c.

10.3 The pressure level

This mode uses **no external reference pressure**, neither at the bottom nor anywhere else. The pressure on the cut level is not an unknown either: it keeps its value from the starting state. That value comes from integrating the hydrostatic equation in τ, $dP/d\tau = g/\kappa$, from the top of the grid, where $P_1 = g\tau_1/\kappa_1$ (the gas above the top level assumed to have the opacity κ_1 and zero pressure at $\tau = 0$). Because τ measures opacity-weighted mass, this fixes the pressure without any outside information: at the cut level the pressure is the weight of the gas that produces the optical depth 10^{-4}. Below the cut the vertical residual R_z links each pressure to the one above it, starting from this fixed value; the pressures below are thus set by the force balance and the τ scale. The price is that the result depends on the opacity: if the τ scale of the input was computed with a different opacity, the pressures differ in proportion.

10.4 The starting state

The iteration starts from the column-hydrostatic solution in τ: each column integrated from the top with $d\ln P/d\ln\tau = g\tau\rho/(\chi P)$ and $dz/d\ln\tau = -\tau/\chi$ (Heun substeps, $\Delta\ln P \le 0.05$; the rise of $\ln P$ limited to 5 per 0.1 dex against an interpolation artefact of hot-over-cool layers), aligned so that the total pressure $P + B^2/2\mu_0$ is the same in all columns at one height, and gauged so that the mean height of $\tau = 1$ is zero. From this state come the fixed upper levels, the cut-level pressures, the weights of Φ, and the first guess of u and c.

11. Assumptions

A1. Static gas: velocities neglected.
A2. T and B fixed on the τ levels. The field values travel with their levels; the Lorentz force is computed from them at the current heights.
A3. Best compromise, not equilibrium: the solution minimises Eq. (13).
A4. Pressure level from the τ scale: the pressure on the cut level is that of the column-hydrostatic solution (top closure $P_1 = g\tau_1/\kappa_1$); no external reference pressure.
A5. Fixed upper atmosphere: above $\log_{10}\tau = -4$ the columns are τ-hydrostatic and rigid.
A6. Consistent opacity: the input τ scale is the vertical 500 nm continuum optical depth of the same LTE equation of state and opacity as the table.
A7. Horizontally periodic box, uniform horizontal spacing.
A8. Weights from the starting state, fixed.
A9. LTE ideal gas; constant vertical gravity; plane-parallel geometry.
A10. No condition on $\nabla\cdot\mathbf{B}$: nothing enforces $\nabla\cdot\mathbf{B} = 0$ for the field at its new positions.

12. The discrete problem

For each column (i, j) and solved level $k = k_c \ldots N_\tau$:

1. **State:** $q_{ijk} = \rho/P$, $\ln\chi_{ijk}$ and $a_{ijk} = \exp(s_k - \ln\chi_{ijk})$ from the table at $(\ln T_{ijk}, u_{ijk})$.
2. **Heights:** $z_{ij,k_c} = c_{ij}$; levels above the cut at c_{ij} plus their fixed offsets; levels below by Eq. (3).
3. **Field derivatives** on all levels: along the column $(\partial_z f)_{ijk} = (f_{ij,k-1} - f_{ij,k+1})/(z_{ij,k-1} - z_{ij,k+1})$ (one-sided at the top and bottom levels); on the τ surface centred periodic differences, $(\partial_x f)|_\tau = (f_{i+1,jk} - f_{i-1,jk})/(2\Delta x)$; then Eq. (11) with the surface slopes $(\partial_x z)|_\tau$, $(\partial_y z)|_\tau$ computed the same way. Then $\mathbf{J}$, $\mathbf{L}$.
4. **Residuals:** with $W_{ijk} = g\,q_{ijk} - (L_z)_{ijk}e^{-u_{ijk}}$,

$$(R_x)_{ijk} = -\frac{u_{i+1,jk} - u_{i-1,jk}}{2\Delta x} - W_{ijk}\,\frac{z_{i+1,jk} - z_{i-1,jk}}{2\Delta x} + (L_x)_{ijk}e^{-u_{ijk}}, \tag{14}$$

$$(R_z)_{ij,k+1/2} = u_{ij,k+1} - u_{ijk} - \frac{\Delta s_k}{2}\big(W_{ijk}a_{ijk} + W_{ij,k+1}a_{ij,k+1}\big), \tag{15}$$

and R_y like R_x. The vertical residual is the trapezoidal integral of Eq. (8) over the interval, the same rule that gives the heights in Eq. (3).

13. The iteration

13.1 Gauss–Newton

Φ depends non-linearly on the unknowns $v = (c, u)$ through the equation of state, the opacity in the heights, and the field derivatives. Each step replaces the residual vector by its linear approximation, $\mathbf{R}(v + \delta v) \approx \mathbf{R} + \mathsf{J}\delta v$, and minimises the resulting quadratic (Gauss–Newton, the least-squares form of Newton's method). Setting its gradient to zero gives the normal equations

$$\boxed{\big(\mathsf{J}^\top\mathsf{W}\mathsf{J} + \mathsf{E} + \mu_{\mathrm{LM}}\,\mathsf{D}\big)\,\delta v = -\mathsf{J}^\top\mathsf{W}\,\mathbf{R}} \tag{16}$$

with W the diagonal matrix of the weights of Φ. $\mathsf{E} = 10^{-2}\mathsf{D}$ is a small fixed damping and D is diagonal: $\kappa_1^2\,\overline{\Delta z}_k \bar{H}_k^2$ for a pressure unknown and $\kappa_1^2 \sum_k \overline{\Delta z}_k \bar{H}_k^2 \overline{W}_k^{\,2}$ for a column height, with $\kappa_1 = 2\pi/(N_x \Delta x)$ the smallest horizontal wavenumber and $\overline{W}_k$ the layer median of W; this is one per cent of the horizontal stiffness of the longest wave. The damping matters in particular for the uniform shift of all columns, which changes no residual (Sect. 14).

13.2 The exact linearisation

J is the exact derivative of the discrete residuals; it is never stored, only applied to vectors. Its ingredients are:

1. **Equation of state:** $\partial q/\partial u = q(\partial \ln \rho/\partial \ln P - 1)$ from the table derivative, and $\chi' = \partial \ln \chi/\partial \ln P$ by a central difference ($\pm 10^{-4}$ in u). Hence $\partial W/\partial u = g\,\partial q/\partial u + L_z e^{-u}$ and $\partial(Wa)/\partial u = a\,\partial W/\partial u - W\,a\,\chi'$.
2. **Heights:** from Eq. (3), $\delta a = -a\chi' \delta u$, so
$$\delta z_{k_c} = \delta c, \qquad \delta z_{k+1} = \delta z_k + \frac{\Delta s_k}{2}\big(a_k \chi'_k\,\delta u_k + a_{k+1}\chi'_{k+1}\,\delta u_{k+1}\big), \tag{17}$$
a cumulative sum down the column (with $\delta u_{k_c} = 0$), and $\delta z = \delta c$ above the cut.
3. **Lorentz force through the heights:** a change of the heights changes the vertical spacing, $\zeta_k = (\delta z_{k-1} - \delta z_{k+1})/(z_{k-1} - z_{k+1})$, which changes $\partial_z f$ by $-\zeta_k\,\partial_z f$; and it changes the slopes by $\partial_x \delta z$, $\partial_y \delta z$, which changes $\partial_x f|_z$ by $-(\partial_x \delta z)\,\partial_z f - (\partial_x z)\,\delta(\partial_z f)$. The resulting $\delta\mathbf{L}$ is linear in ζ, $\partial_x \delta z$ and $\partial_y \delta z$ and enters all three residuals (R_x, R_y directly and through W; R_z through W).
4. **Direct terms:** $-\partial_x \delta u$, $-W\,\partial_x \delta z$, $-(\partial_x z)\,\partial W/\partial u\,\delta u$ and $-L_x e^{-u}\delta u$ in R_x (likewise R_y); $\delta u_{k+1} - \delta u_k - \frac{\Delta s}{2}\sum \partial(Wa)/\partial u\,\delta u$ in R_z.

Its transpose $\mathsf{J}^\top$ is applied by reversing each step (the cumulative sum becomes a sum from the bottom up). Both have been checked against finite differences of the residuals.

13.3 Solving the linear system

Equation (16) is symmetric and positive definite and is solved by the preconditioned conjugate-gradient method (only matrix–vector products are needed), from $\delta v = 0$, to a relative residual of 10^{-3} or at most 80 iterations.

13.4 The preconditioner

In a horizontally periodic box any function is a sum of horizontal waves, and a centred difference acts on a wave (k_x, k_y) as multiplication by $i \sin(k_x \Delta x)/\Delta x$. If all coefficients were the same throughout each level, the operator would act on each wave separately as an $n \times n$ matrix
$$\mathsf{A} + \sigma^2 \mathsf{B}, \qquad \sigma^2 = \frac{\sin^2(k_x \Delta x)}{\Delta x^2} + \frac{\sin^2(k_y \Delta y)}{\Delta y^2}, \tag{18}$$
where $\mathsf{A} = \mathsf{G}_z^\top \mathsf{W}_z \mathsf{G}_z + \mathsf{E} + \mu_{\mathrm{LM}}\mathsf{D}$ comes from the vertical residuals (G_z the vertical Jacobian of one column) and $\mathsf{B} = \mathsf{G}_h^\top \mathsf{W}_h \mathsf{G}_h$ from the horizontal ones, $\mathsf{G}_h = -\mathsf{I}_u - \overline{W}\,\mathsf{S}$ with S the cumulative height operator of item 2 above. The preconditioner builds A and B from the layer medians of $a\chi'$, $\partial(Wa)/\partial u$ and W (the Lorentz-through-height and local Lorentz terms are left out). Rather than factorising $\mathsf{A} + \sigma^2\mathsf{B}$ for each of the $N_x N_y$ waves, it is diagonalised once: with the Cholesky factor $\mathsf{A} = \mathsf{L}\mathsf{L}^\top$ and the eigenvectors V and eigenvalues $\lambda_m \geq 0$ of $\mathsf{L}^{-1}\mathsf{B}\mathsf{L}^{-\top}$, the matrix $\mathsf{Q} = \mathsf{V}^\top \mathsf{L}^{-1}$ satisfies $\mathsf{Q}\mathsf{A}\mathsf{Q}^\top = \mathsf{I}$ and $\mathsf{Q}\mathsf{B}\mathsf{Q}^\top = \mathrm{diag}(\lambda)$, so
$$(\mathsf{A} + \sigma^2\mathsf{B})^{-1} = \mathsf{Q}^\top \,\mathrm{diag}\!\left(\frac{1}{1 + \sigma^2 \lambda_m}\right) \mathsf{Q} \tag{19}$$
for every wave at once. Applying the preconditioner costs a horizontal Fourier transform, two products with the small matrix Q, and an inverse transform.

13.5 Step control: Levenberg–Marquardt

A pure Gauss–Newton step can be too ambitious where the problem is strongly non-linear. Three safeguards keep it reliable:

1. **Whole-step scaling.** With $s_{\mathrm{c}} = \min\big(1,\ 2/\max|\delta u|,\ 300\,\mathrm{km}/\max|\delta c|\big)$, the first trial is $\alpha = s_{\mathrm{c}}$: no pressure changes by more than a factor e^2 and no column moves by more than 300 km (`dc_cap_m`). Scaling the whole step keeps its direction, which is a descent direction of Φ; clamping individual components would not guarantee that.
2. **Backtracking:** $\alpha = s_{\mathrm{c}}, s_{\mathrm{c}}/2, s_{\mathrm{c}}/4, s_{\mathrm{c}}/8$; the first trial that lowers Φ and stays inside the equation-of-state table is accepted.
3. **Damping:** if none of the four trials is accepted, the damping is raised, $\mu_{\mathrm{LM}} \leftarrow \max(4\mu_{\mathrm{LM}}, 1)$, and the step recomputed (at most 8 times). Large μ_{LM} turns the step from the Gauss–Newton direction toward the direction of steepest descent with a short length. After an accepted step μ_{LM} is divided by 3 if the full, unscaled step was taken ($\alpha = 1$), kept if $\alpha \geq s_{\mathrm{c}}/2$, doubled otherwise, and set to 0 once below 10^{-3}. It starts at `lm_mu0` $= 0$ (pure Gauss–Newton).

13.6 Stopping rule

After each accepted step the relative decrease $\Delta\Phi/\Phi = (\Phi_{\text{old}} - \Phi_{\text{new}})/\Phi_{\text{old}}$ is computed. The iteration stops when

- $\Delta\Phi/\Phi < \texttt{rtol} = 10^{-4}$ on $\texttt{n_consec} = 3$ consecutive steps (converged); or
- the number of steps reaches `max_steps` (100); or
- no step is accepted even with the largest damping.

The criterion is on the objective only. A single small decrease is not enough, because Φ is nearly flat along the heights of a few strong-field columns (Sect. 14) and an isolated small step can occur long before the minimum. The largest height change of each step and the percentiles of $z(\tau = 1)$ are logged but not used to stop.

13.7 After the iteration

All columns are shifted together so that the mean height of $\tau_{500} = 1$ is zero (the gauge), and ρ is evaluated on all levels.

14. What is determined, and what is not

- **The global height.** Moving all columns up or down together changes no residual: the forces depend only on differences and slopes of the heights. This offset is fixed by the gauge, a convention, not a measurement.
- **The relative heights** of the columns are determined by the horizontal balance through the tilt term of Eq. (10), not by the alignment of the starting state, which serves only as the first guess. A few strong-field columns are weakly constrained: their heights can change noticeably while Φ barely changes.
- **The pressure level** comes from the τ scale (Sect. 10.3).
- **Above the cut** nothing is solved; the levels keep the hydrostatic structure of the starting state.
- **The magnetic field is not made consistent with the new geometry.** The field values on the τ levels are kept while the levels move in space; $\nabla \cdot \mathbf{B} = 0$ is not enforced for the field at its new positions. The Lorentz force is computed from the field as it lies.

15. What decides the heights: the Wilson depression

Consider a column with a strong, nearly vertical field among weak-field neighbours. Its magnetic pressure $B^2/2\mu_0$ pushes outward on the neighbours, so at the same height it can hold its own against them with less gas pressure. Less gas pressure means lower density and lower opacity, so light escapes from deeper: the τ surfaces of this column lie lower. Horizontal balance favours exactly this through the tilt term of Eq. (10). The depression of strong-field regions, seen in sunspots near the limb as the **Wilson depression**, is therefore expected to come out of the force balance rather than being put in by hand, although for a given column the result also depends on its temperature structure and on how well the compromise balances the forces there. Conversely, hot granules have a high opacity, and their τ surfaces tend to lie higher.

16. Summary of the algorithm

1. Read ℓ_k, x, y, T, $\mathbf{B}$. Compute the column-hydrostatic starting state in τ (top closure, Heun integration, limited pressure rise, total-pressure alignment, gauge).
2. Choose the cut level nearest $\log_{10}\tau = -4$. Fix the levels above it relative to the cut level, and the cut-level pressure. Unknowns: c per column and $\ln P$ below the cut.
3. Compute the weights $\bar{H}_k$, $\overline{\Delta z}$ from the starting state; evaluate heights, $\mathbf{L}$, residuals and Φ.
4. Repeat: build and diagonalise the layer-median preconditioner; solve the damped normal equations by preconditioned CG; scale the step as a whole to the caps; backtrack; raise the damping if no trial lowers Φ; accept and update the damping. Stop after three consecutive steps with $\Delta\Phi/\Phi < 10^{-4}$, after 100 steps, or when no step is accepted.
5. Gauge the mean height of $\tau_{500} = 1$ to zero; compute ρ; write z, P, ρ, T, $\mathbf{B}$ and $z(\tau = 1)$.